MANAGEMENT & TEAM BRIEFING

Inventory Forecasting

From daily transactions to trusted, human-controlled inventory decisions

01 **CAPTURE**
Sales, receipts, returns, transfers

02 **UNDERSTAND**
Ledger, FIFO, analytics, snapshots

03 **PREDICT**
Baselines + trained neural models

04 **DECIDE**
Buy, reduce, stop, transfer, clear

10 phases delivered

Human approval retained

Production controls still required

03 September 2026

EXECUTIVE SUMMARY

A single operating system for stock decisions

One source of operational truth, translated into measurable action without removing management control.

1 OBSERVE

Multi-warehouse balances
Append-only stock ledger
FIFO batches and costs
Purchasing, sales and returns

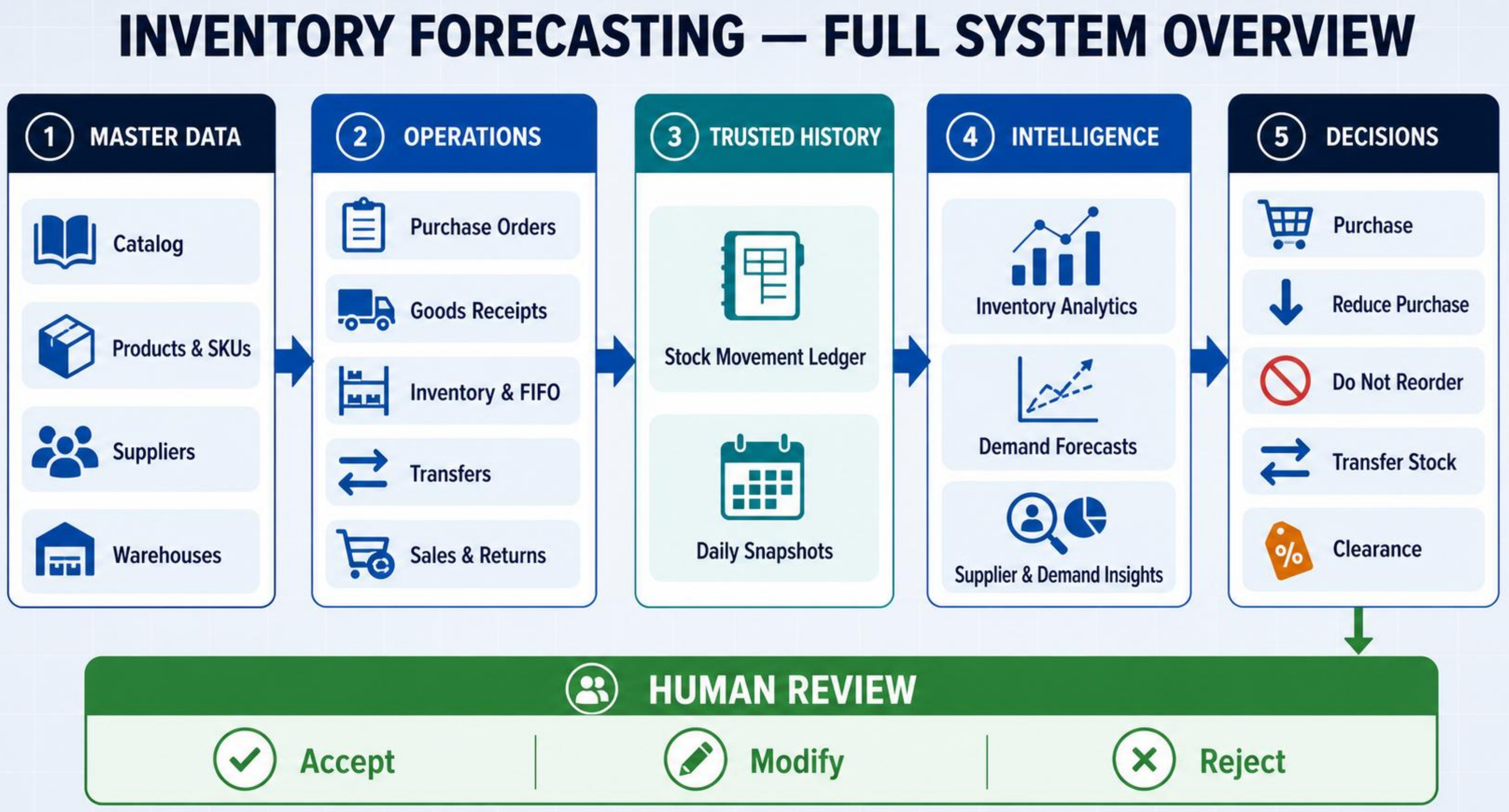
2 UNDERSTAND

Inventory health analytics
Daily stock and demand history
Forecasts with uncertainty
Seven intelligence views

3 ACT

Purchase or reduce buying
Do not reorder or clear
Transfer between warehouses
Accept, modify or reject

Management truth: **Functionally complete, but not production-ready until access roles, registration policy, hosting, backups, mail, workers, scheduler and monitoring are controlled.**



INVENTORY FORECASTING — FULL SYSTEM OVERVIEW
1 MASTER DATA
Catalog
Products & SKUs
Suppliers
Warehouses
2 OPERATIONS
Purchase Orders
Goods Receipts
Inventory & FIFO
Transfers
Sales & Returns
3 TRUSTED HISTORY
Stock Movement Ledger
Daily Snapshots
4 INTELLIGENCE
Inventory Analytics
Demand Forecasts
Supplier & Demand Insights
5 DECISIONS
Purchase
Reduce Purchase
Do Not Reorder
Transfer Stock
Clearance
HUMAN REVIEW
Accept
Modify
Reject


FROM DAILY OPERATIONS TO MANAGEMENT ACTION
1 SET UP MASTER DATA
Products, SKUs, suppliers, warehouses
2 BUY STOCK
PO
Create PO → Approve → Order
3 RECEIVE STOCK
Goods receipt → Inventory & FIFO
4 MANAGE STOCK
Adjustments and warehouse transfers
5 SELL & RETURN
Confirm sale → Record return
6 BUILD TRUSTED HISTORY
Movement ledger → Daily snapshots
7 FORECAST DEMAND
Queue run → Predict → Score accuracy
8 RECOMMEND ACTION
Buy, reduce, stop, transfer, clear
9 HUMAN DECISION
Accept → Modify → Reject
EXECUTE MANUALLY
Create a purchase order or stock transfer

CAPABILITY MAP

What the application contains

Six connected work areas cover the full inventory lifecycle and the intelligence built on top of it.

CAT Catalog

Categories, brands, attributes
Products, variants and SKUs

INV Inventory

Warehouses, balances, movements
Adjustments, FIFO, transfers, snapshots

BUY Purchasing

Suppliers and SKU terms
Purchase orders and goods receipts

SELL Sales

Draft and confirmed sales orders
Sellable and damaged returns

FCST Forecasting & decisions

Runs, forecasts and accuracy
Recommendations and central allocation

AI Advanced intelligence

Supplier, lost sales, anomalies
Relationships, elasticity, promotions

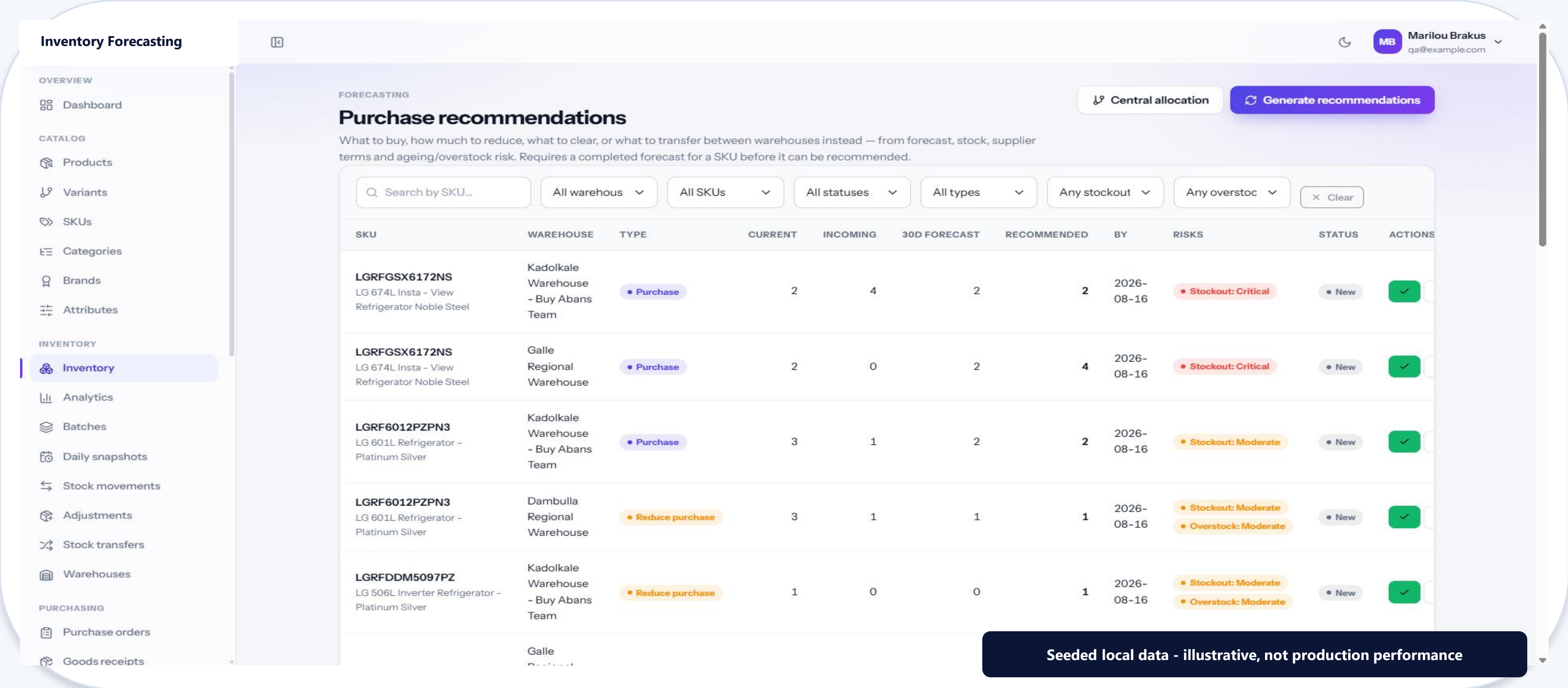
All business routes require authenticated + verified users

Role-based restrictions are not yet implemented

REAL APPLICATION VIEW

The decision workspace users see

Recommendations combine forecast, stock, supplier terms and risk into an auditable human decision.

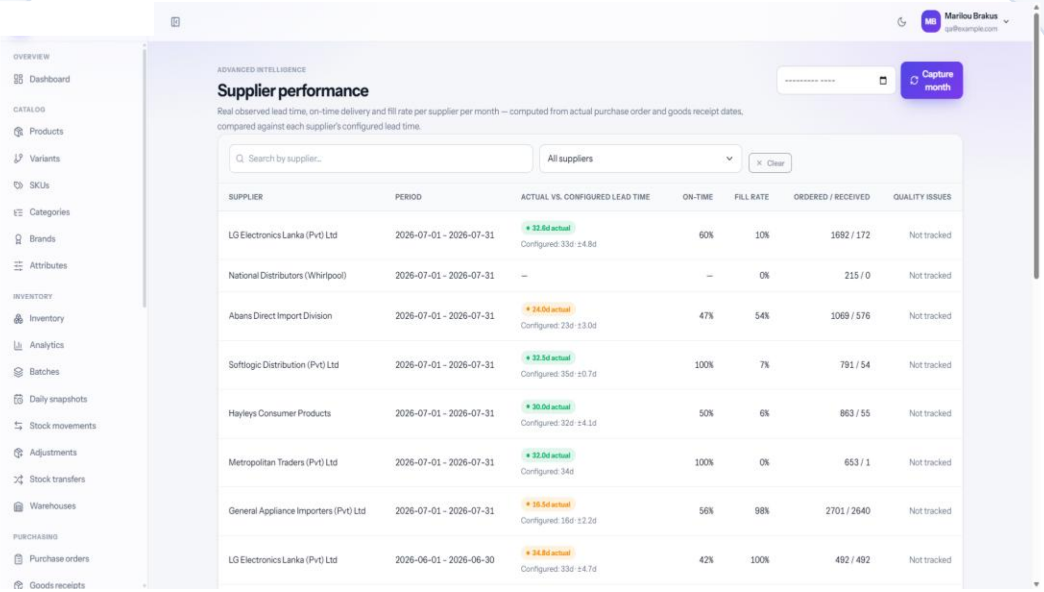


Seeded local data - illustrative, not production performance

REAL APPLICATION VIEWS

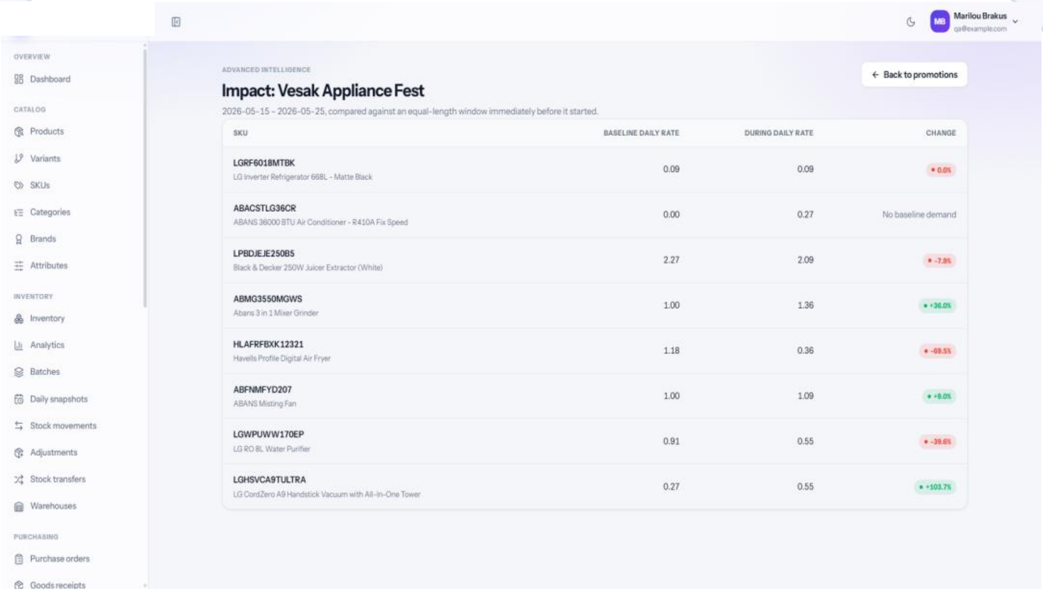
Intelligence remains traceable to operational data

Supplier scorecards and promotion impact are examples of the seven intelligence views delivered.



Supplier performance

Actual lead time, on-time rate and fill rate from PO/receipt dates



Promotion impact

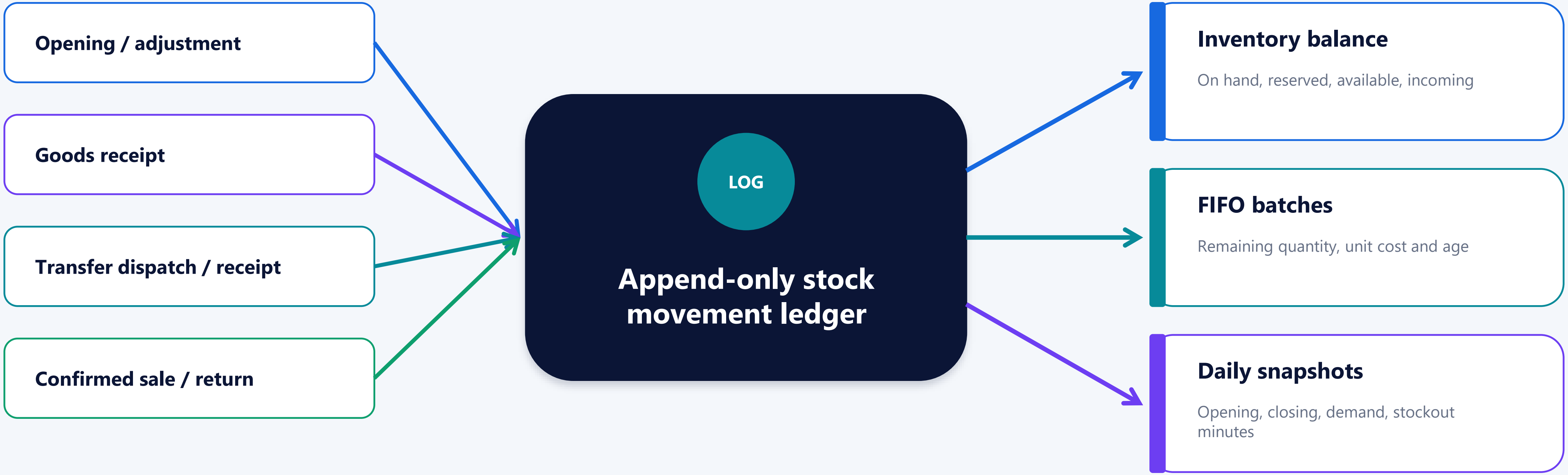
Before-vs-during demand change per promoted SKU

Evidence-led signals, not autonomous actions

PROCESS INFOGRAPHIC

Every stock change follows one controlled path

Balances are derived from an append-only movement ledger; users never edit stock totals directly.



Transactional updates | **Negative stock blocked** | **Weighted average + FIFO cost**

STATUS LIFECYCLES

Operational workflows protect stock integrity

Status changes are business controls: they define when stock and financial values become final.

PURCHASE ORDER

Draft → Approved → Ordered → Partially received → Received

Only Draft can be edited or deleted. Cancel before stock arrives.

STOCK TRANSFER

Draft → Approved → Dispatched → Received

Source decreases at dispatch; destination increases at receipt.

SALES ORDER

Draft → Confirmed → Return if needed

Confirmed sale consumes FIFO; damaged returns are logged in and out.

DETERMINISTIC ANALYTICS

Analytics turns inventory into management questions

Calculated on request from stock and sales history - read-only, transparent and independent of ML.

V **Daily velocity**

How quickly does it sell?

D **Days of stock**

How long until run-out?

T **Turnover**

How productively is stock used?

A **Stock age**

How long has cash been sitting?

ABC **Revenue tier**

Which items matter most to revenue?

SPD **Movement class**

Fast, slow or dead mover?

ROP **Reorder point**

Will available stock cover lead time + safety?

Filters: warehouse | category | SKU | ABC | mover | reorder need | lookback | safety days

PROCESS INFOGRAPHIC

How the forecast is produced today

The system chooses the demand source and algorithm per SKU, records what actually ran, and scores it after the horizon closes.

1 INPUTS

Daily demand
Stockouts
Price & promotion
SKU metadata

2 MATURITY

Cold start
Early
Established
Mature / declining

3 SOURCE

Own history
Category + size
Brand + category
Category / hybrid

4 ALGORITHM

EWMA
Seasonal naive
TFT
DeepAR

5 OUTPUT

Quantity
Lower / upper
Confidence
Actual algorithm

6 LEARN

Wait for horizon
Compare actuals
Score accuracy
Rank next run

Default: EWMA until scored accuracy history exists

Neural refusal -> safe baseline fallback, per SKU

MODEL EVIDENCE & GOVERNANCE

The model is operational; its business accuracy is not yet proven

Lower WAPE is better. These results are pipeline evidence from generated history, not a production forecast claim.

Most recent honest 30-day window

EWMA currently wins this checkpoint window.

What is working now

441
warehouse / SKU pairs processed

404
TFT forecasts in a real seeded run

37
safe baseline fallbacks

~20s
warm end-to-end neural run

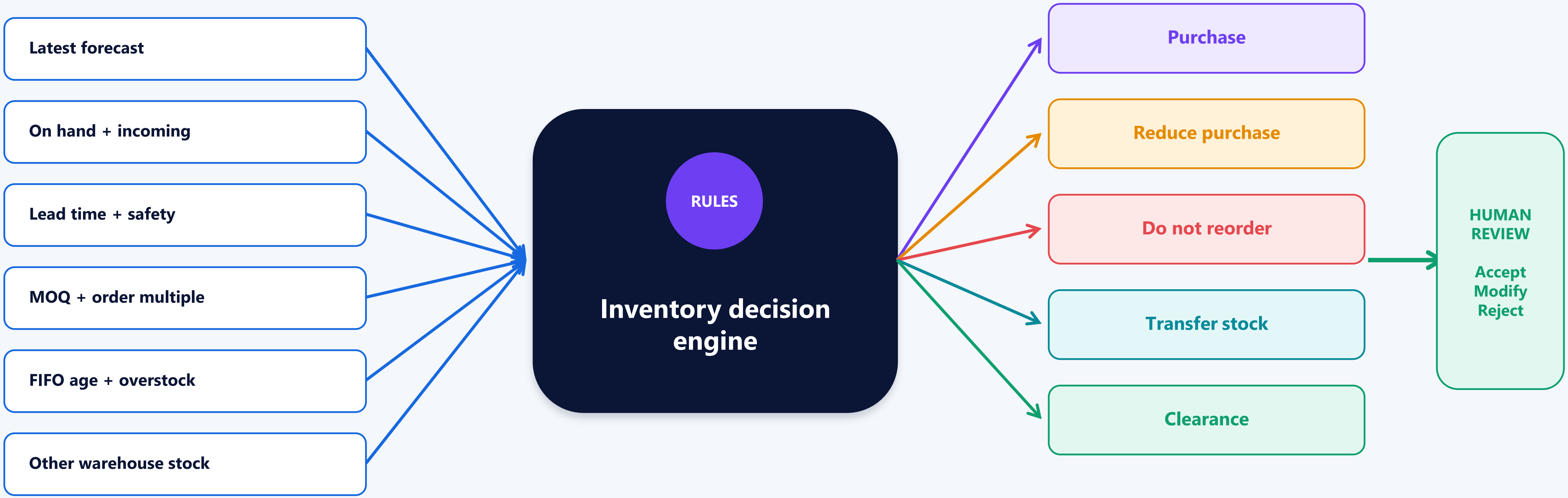
Decision: keep EWMA as default until real sales history validates a neural promotion.

A fresh TFT led the best baseline by about 7.8% in the first three rolling windows - but on synthetic history.

PROCESS INFOGRAPHIC

A forecast becomes one of five controlled actions

The decision engine combines demand, supply, ordering constraints, ageing and location before asking a person to decide.



Accepted decisions are executed through normal PO or transfer workflows - never silently auto-posted.

MULTI-LOCATION DECISION FLOW

Optimize across warehouses before buying more

Stock transfer matching reduces avoidable purchasing; central allocation consolidates the remaining need.

Warehouse A

+42 surplus

Warehouse B

-18 shortage

Warehouse C

-12 shortage

TRANSFER MATCHING

Use another warehouse first when one source can cover the recipient's full need.

No route or transport-cost optimization yet.

BUY

CENTRAL ALLOCATION

For SKUs still needed in two or more warehouses:

- combine the quantity
- retain warehouse breakdown
- place a consolidated supplier order

Business effect: rebalance owned stock before committing more working capital

ADVANCED INTELLIGENCE

Seven additional signals expand the management picture

Each view is independent, explainable and computed from operational history.

SUP
Supplier performance
Actual lead time, on-time delivery and fill rate

LOST
Lost-sales estimate
Likely missed units during real stockouts

ANOM
Demand anomalies
Statistical demand outliers by SKU and date

SIM
Product similarity
Comparable items from product attributes

REL
Cannibalization & successor
Products that move against or replace each other

ELAS
Price elasticity
Demand response across actual price points

PROMO
Promotion impact
Before-vs-during demand by promotion and SKU

Interpret low-sample signals carefully: availability is not the same as statistical certainty

PROCESS INFOGRAPHIC

Automation keeps history, accuracy and decisions current

The queue worker and scheduler are operational dependencies; without them, the application appears healthy while intelligence becomes stale.

EVERY DAY

00:15

Capture snapshots

Previous day balances, demand and stockout minutes

00:30

Score due forecasts

Compare elapsed horizons with actual demand

00:45

Generate recommendations

Refresh purchase, transfer and ageing actions

FIRST DAY OF EACH MONTH

01:00 - Capture supplier performance

02:00 - Retrain neural models in background

Manual buttons remain available for controlled backfill and on-demand runs

TECHNICAL ARCHITECTURE

The system is one product across two runtimes

Laravel owns business truth and workflow; Python serves statistical and neural forecasts through a narrow HTTP contract.

USERS

Browser
Login + verification
2FA + passkeys

WEB APPLICATION

Inertia v3 + React 19
Bootstrap 5.3 + SCSS
Typed Wayfinder routes

LARAVEL 13

Form Request -> Controller
Facade -> Service -> Model
Transactions + validation

MYSQL

Operational tables
Ledger + snapshots
Forecasts + decisions

PYTHON

FastAPI
EWMA / seasonal
TFT / DeepAR

Database queue worker: forecast jobs

Scheduler: daily + monthly sequence

Python is separate: supervise it and require bearer-token auth before network exposure

Controls delivered - and controls still required

The application protects individual accounts and stock transactions, but does not yet separate business duties by role.

DELIVERED CONTROLS

- OK Verified login, reset and password confirmation
- OK TOTP two-factor authentication and passkeys
- OK Login / 2FA / passkey rate limits
- OK Append-only ledger and immutable posted documents
- OK Transactional stock writes and negative-stock guard
- OK Server-calculated PO totals and FIFO sale cost

BEFORE PRODUCTION

- ! Role-based permissions and separation of duties
- ! Controlled registration / user provisioning policy
- ! Real email, HTTPS and production cookie settings
- ! Hosting, backup, restore and monitoring design
- ! Supervised queue, scheduler and Python processes
- ! Token-protect the Python service off localhost

ASSURANCE SNAPSHOT

Delivery evidence shows breadth and working integration

Counts describe the current source tree and the latest fully logged end-to-end verification.

10	delivery phases implemented
254/254	Pest tests passed
62/62	Python tests passed
31	Facade + service module pairs
33	Eloquent models
38	database migrations
71	React page files
197	recommendations generated in seeded run

Latest end-to-end seeded run: 441 pairs -> 404 TFT forecasts + 37 safe fallbacks

Known UI gap: dashboard KPI props remain unconnected and show honest empty states

WHAT IS DELIVERED — WHAT IS NEEDED BEFORE GO-LIVE

DELIVERED CONTROLS

- ✓ Verified login
- ✓ Immutable stock ledger
- ✓ Negative-stock guard
- ✓ FIFO costing
- ✓ Transactional stock updates
- ✓ Daily snapshots
- ✓ Forecast accuracy scoring
- ✓ Human decision workflow

BEFORE GO-LIVE

- ! Role-based permissions
- ! Controlled registration
- ! Production hosting & backups
- ! Real email service
- ! Queue supervision & scheduler
- ! Private token-protected Python service
- ! Monitoring & alerts
- ! Full release and browser validation

FUNCTIONALLY COMPLETE → **NOT YET PRODUCTION READY**

Management ownership turns software into an operating capability

Use the system on a cadence, assign decision owners, and promote forecasting models only when real evidence supports it.

DAILY

Review stockout / ageing exceptions
Check queued or failed forecast runs
Post receipts, sales, returns and transfers

WEEKLY

Approve / modify recommendations
Review open and partially received POs
Investigate lost sales and anomalies

MONTHLY

Review supplier scorecards
Review forecast accuracy and model choice
Evaluate promotions and data quality

DECISIONS MANAGEMENT SHOULD MAKE NOW

1. Name data owners, transaction operators and recommendation approvers
2. Implement RBAC and decide how users are provisioned
3. Choose production hosting, MySQL/Redis, backups, mail and monitoring
4. Operate queue, scheduler, Python service and monthly retraining
5. **Validate on real sales history before making TFT or DeepAR the default**

8-MINUTE LIVE DEMO ROUTE

Show the closed loop, not every menu item

1 Products

Show catalog structure and SKU ownership

2 Inventory

Show balance, incoming stock and average cost

3 Stock movements

Prove the append-only audit trail

4 Forecasts

Show source, range, confidence and algorithm

5 Recommendations

Show risks and Accept / Modify / Reject

6 Supplier or promotion insight

Close with a management signal

The value is the closed loop:

transactions become trusted history -> history becomes forecasts -> forecasts become controlled action.